

Modeling and Clustering Cryptocurrency Dynamics Using Machine Learning

This project investigates cryptocurrency price behavior and volatility using machine learning models and advanced exploratory data analysis (EDA). The dataset, obtained from Kaggle (*Every Cryptocurrency Daily Market Price*, 2018), includes daily open, high, low, close, volume, and market capitalization data for over 2,000 cryptocurrencies from 2013 onward. To maintain interpretability and reduce computational complexity, the scope of analysis was narrowed to the top 10 coins by average market capitalization.

Initial preprocessing involved filtering by symbol and sorting by both coin and date to ensure temporal consistency. Several domain-specific features were engineered, such as daily returns, lagged close values, price spreads, rolling volatility (3-day and 7-day), rolling means, and momentum indicators like open-close difference. Outliers like Bitcoin (BTC) were analyzed separately due to their market dominance. Visualizations using Plotly illustrated differences in daily returns, trading volumes, and volatility across assets. Correlation matrices and pair plots revealed strong multicollinearity among some indicators, which informed later feature selection.

For supervised modeling, the task was to classify whether Bitcoin's price would rise the following day. Two models were tested: a Decision Tree classifier and a Random Forest classifier, both trained using a balanced target variable. The Random Forest performed better, achieving higher accuracy and F1 scores. The analysis of the feature importance highlighted that volatility and short-term momentum indicators were among the most predictive attributes.

The second task involved unsupervised learning through K-Means clustering to identify groups of coins with similar return and volatility profiles. Feature standardization was performed prior to clustering. The model was able to identify clusters of coins exhibiting similar behavior, offering insights into potential strategies. Interestingly, BTC (Bitcoin) showed relatively low short-term returns during this period, suggesting that risk-adjusted performance may favor alternative coins. Sharpe and Sortino ratios were calculated to evaluate average return against volatility and downside risk.

This demonstrates how combining domain knowledge, feature engineering, and classical machine learning models can yield actionable insights into a complex and rapidly evolving financial market. Future research could involve using neural networks for modeling or incorporating external signals such as macroeconomic trends, blockchain metrics, or social sentiment and using natural language processing in the future.

References

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